UNIVERSITY OF SRI JAYEWARDENEPURA - FACULTY OF APPLIED SCIENCES

BSc General Degree Second Year First Semester Course Unit Examination – September/October 2025

DEPARTMENT OF PHYSICS

PHY201 2.0 Optics 1

Time: Two (02) hours	**No. of questions:** 04	**No. of pages:** 03	**Total marks: 100**

Answer ALL questions

1)

a) Clearly explaining the geometry of the **given curvatures** and principles related to propagation of light explain the following.

i. From a source at one focus of an **ellipse**, the image forms at the other focus.

ii. From a source at infinity, the image form at the focus of a **parabola.**

[05 marks]

b) The Figure 1A shows a graphical construction made to evaluate the direction of a refracted ray off an interface with refractive indices *n* and *m*. The line OA represents the direction of the incident ray, while the line OR represents the direction of the refracted ray and a line NN′ represents the normal to the interface. A similar diagram as of Figure 1A can be used to evaluate the direction of a reflected ray.

i. **Draw** the diagram for the reflection using above representations. **[05 marks]**

ii. **Show** that above representation satisfies the Law of reflection. **[05 marks]**

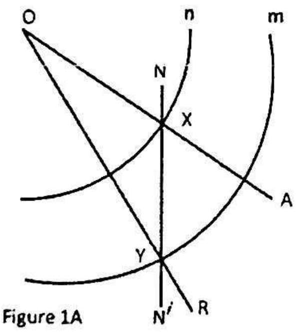
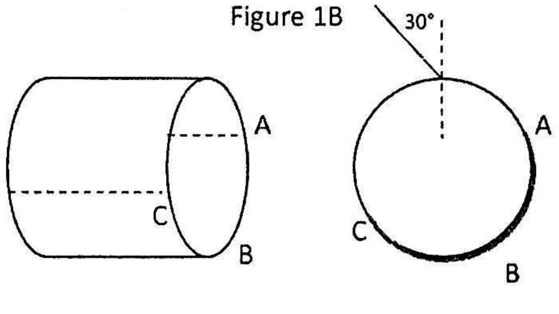


Figure 1A

Figure 1B



c) As shown in Figure 1B, a light ray aimed towards the topmost point of the surface of a glass (refractive index 1.5) cylinder of radius 5.0 cm makes an incident angle of 30°. The figure is not to scale. A half of the spherical surface of this rod (the side labeled as ABC) is silvered on its length so the light will reflect off that surface. **Show** the path of the ray passing through and exiting the cylinder.

[10 marks]

[25 marks]

2)

a) Consider a concave mirror, radius of curvature R. Assume an object with height h with object distance s $(2R > s > R)$. The image distance is s' and the image height is h'. Draw a clear ray diagram assuming paraxial rays. **[05 marks]**

b) Using the geometry in the above ray diagram derive the mirror equation. **[05 marks]**

c) Consider a lens system consist of two convex lenses. Focal length of each is f and the two lenses are separated by a distance $2f$. An object (O) is placed at a distance s $(2f > s > f)$ in front of the objective lenses. Determine the image position (I) using the oblique ray method. **[05 marks]**

d) Some neutron stars emit harmful radiation concentrated in specific directions, which can wipe-out life on planets. Assume these radiation acts like how visible light acts passing through different media. Consider such star at 80 light-years (ly) from earth and in-between the neutron star and the earth following condition exist. At 10 ly from the star there is a media with refractive index n_1 ($n_1 = 1.25$) and light enters in to this media through a flat interface and exits through a convex surface of 15 ly in radius of curvature. The thickness of this media is 25 ly. After passing the above region, at 10 ly distance, it interacts with a region concentrated with dark matter before finally reaching the region of our solar system. The dark matter acts like a convex lens of focal length 10 ly. Refractive index of the empty space (vacuum) is 1.0.

i. Determine the transfer-ray-matrix for the above system. **[07 marks]**

ii. If the initial ray emits at 10° above the line of earth and the star, will the life on earth survive from this blast of radiation. **[03 marks]**

For REPEAT STUDENTs:

i. Derive the formula for apparent depth h' for an object in a media of refractive index n_2 at a real depth h, observing through a medial of refractive index n_1. **[04 marks]**

ii. Find the apparent depth of the above neutron star, observing from earth. **[06 marks]**

[25 marks]

3)

a) Describe the Young's double slit experiment using a suitable diagram. In your answer, clearly indicate the path difference between the two rays and explain how it leads to the observed interference pattern. **[05 marks]**

b) Given the expression for the intensity distribution, $I = 4a^2\cos^2(\delta/2)$, show graphically how the intensity varies with the phase difference between rays from the two coherent sources. **[05 marks]**

c) Draw a well-labeled diagram of a Michelson interferometer, clearly identifying its main components and paths of rays. **[05 marks]**

d) Discuss the nature of the interference fringes observed in a Michelson interferometer when:
i). the two mirrors are perfectly perpendicular to each other.
ii). the two mirrors deviate from the perpendicular positions. **[05 marks]**

e) Derive the expression for Brewster's law in polarization using a suitable diagram.

[05 marks]

[25 marks]

4)

a) Apply Stoke's treatment to a ray reflected and refracted at the boundary separating two media and derive the following results. Notations have their usual meanings.

i). $tt' = 1 - r^2$
ii). $r' = -r$ **[05 marks]**

b) Using a clearly labeled diagram, explain the intensity distribution observed on a screen due to double-slit diffraction. Clearly indicate the positions of the central maximum, subsidiary maxima, and minima in the intensity distribution, and explain the conditions for their occurrence.

Hint: $I = 4A_o^2 \frac{sin^2\beta}{\beta^2} cos^2\gamma$ **[05 marks]**

c) i). State the mathematical expression for the **amplitude** of single-slit diffraction pattern. Describe each parameter in the expression. **[05 marks]**

ii). Derive an expression for the angular position of the first minimum in a single-slit diffraction pattern. The angular position is defined as the angle between the central axis of the slit and the direction of the point under consideration on the diffraction pattern. **[05 marks]**

iii). Monochromatic light of wavelength 750 nm passes through a slit of width 1.0×10^{-3} mm. Calculate the width of central maxima in degrees. **[05 marks]**

[25 marks]

****************************** *End of the Question Paper* **********************